

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY

Department of Computer Science and Engineering

L-4/T-1 CSE 409: Computer Graphics

Time: 25 minutes

Marks: 20

Student Name: _____

Student ID: _____

1. a) What is the physical signification of attenuation of point light? (2)

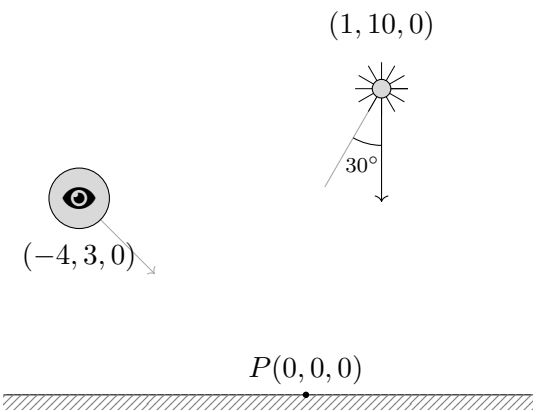
b) Describe how textures can be used to enhance the Phong model. (2)

c) What is the advantage of using the Blinn-Phong model over the Phong model? (2)

2. As a game developer in Iron studio, you are working in a new level of a game. There is just a strange floor at $y = 0$ and a spot light at $(1, 10, 0)$. You have decided the following parameters using the Phong model for the floor:

- Spot light intensity $\mathbf{I} = (10, 10, 10)$
- Falloff exponent $e = 5$
- Ambient light $\mathbf{I}_a = (0.1, 0.1, 0.1)$
- Ambient Coefficient $\mathbf{k}_a = (0.1, 0.2, 0.1)$
- Diffuse Coefficient $\mathbf{k}_d = (0.8, 0.2, 0.6)$
- Specular Coefficient $\mathbf{k}_s = (0.1, 0.6, 0.3)$
- Shininess $n = 10$

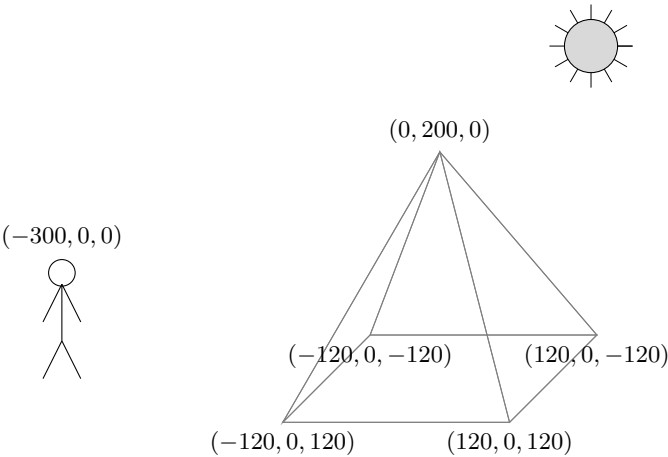
The user begins as a flying soul at $(-4, 3, 0)$ and looks at the floor at $P(0, 0, 0)$.



a) What is the colour at P using the Phong model? (3)

b) What is the colour at P using the Blinn-Phong model? (3)

3. You are visiting the great pyramid of Giza. You are standing 300 meters west at $(-300, 0, 0)$. The pyramid has its base between $(120, 0, 120)$, $(120, 0, -120)$, $(-120, 0, -120)$ and $(-120, 0, 120)$. The top of the pyramid is at $(0, 200, 0)$. It is 10 AM in the morning. The sun is pointing in the direction $-\hat{i} - 2\hat{j}$.



Are you in the shadow of the pyramid? (6)

Hint 1: Consider a shadow ray.

Hint 2: Does the shadow ray intersect the pyramid at all?

4. For your hard work and attending the exam. (2)